Optimal Context Mapping

A Self-Expanding Context Memory with a Routing Law, and Its Token Cost Against Tool-Using Agents

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Abstract. Language-model agents that answer questions over a large document collection spend most of their input tokens re-reading: every round of a tool loop re-sends the instructions, the question and everything returned so far. We study *optimal context mapping*: a context memory (CMLM) that sits beside the model and replaces searching with a short readout of sentences. The memory stores every sentence of the collection as a 1,024-bit random-hyperplane fingerprint of a frozen sentence embedding; a question is answered from the sentences whose fingerprints agree most with it. Three mechanisms are added and tested: (i) *taught shortcuts*, an append-only store of questions the model has answered, which generalise to new wordings through the geometry of the embedding while leaving every stored fingerprint unchanged; (ii) a *routing law*, an 8-bit observation of the question mapped by a verified integer dispatch to one of six readout lanes (certified shortcut, meaning, keyword, mixed, wide, or hand-over to the model's own search); and (iii) *self-expansion*, in which every accepted answer is taught back.

All comparisons are live, pre-registered, and use fresh Claude Opus 5.5 agents graded by blind judges, on 60 held-out questions over nested slices of 20 Newsgroups (1M, 3M and 9M tokens). Against an agent with regex search and file reading, the self-expanding CMLM answered a stream of 120 questions (60 first-time, then the same 60 facts reworded with no shared content word) with equal accuracy (118/120 vs 120/120, 117 vs 117, 117 vs 116; exact McNemar $p \geq 0.375$), 1.10–1.34× fewer input tokens per right answer under a strict per-round pricing (1.5–2.5× when the closing teach rides the answer), and 3.0–5.0× fewer model requests. On the reworded half alone it used 1.7–1.8× fewer tokens and 4.5–9.1× fewer requests. On first-time questions it was not cheaper in tokens than the searching agent. We report the negative results in full: a taught vocabulary of question shapes did not transfer to new questions, a router authored on simulated traces lost to the plain memory on real ones, and the untaught memory's advantage shrank with collection size because of crowding in meaning space, which we measure and attribute partly to fingerprint noise.

Keywords: retrieval-augmented generation; agent memory; locality-sensitive hashing; SimHash; token cost; routing; pre-registration.

1. Introduction

An agent asked a question about a corpus far larger than its context window must search. With ordinary tools (a regex search and a file reader) each round of the loop re-sends the whole conversation, so the input a question costs grows roughly with the square of its number of rounds. The alternative this paper studies is a memory that answers first: before the model does anything, it is handed a short readout of the sentences most likely to answer, and it searches only when that readout fails. If the readout is right often enough, most questions cost one round.

The memory studied here, which we call a CMLM (context memory), is deliberately simple: sentence embeddings compressed to bit fingerprints and compared by counting shared bits. What we add, and measure, is how such a memory should *grow* and *route*. Three questions organise the paper. Can a memory learn from the answers the model finds, so that later questions about the same facts, worded differently, cost almost nothing, without damaging what it already stores? Can a small, verifiable law decide per question which kind of readout to serve? And what does all of this cost, in tokens and in requests, against an agent that simply searches?

Contributions. (1) An append-only teaching rule with an exact integrity property and a measured generalisation to new wordings (§2.3). (2) A routing law over eight free observations, verified exhaustively, tested both on the questions it was meant for and on questions it was not (§2.4, §5.4). (3) A token cost model and a set of live, pre-registered, blind-judged comparisons against tool-using agents, including a self-expanding stream (§3–§5). (4) A measured account of why the untaught memory's advantage does not grow with corpus size (§5.3). (5) A full record of negative results, deviations, and an independent audit (§5.6, §6, Appendix B).

2. System

2.1 From a sentence to a fingerprint

A frozen local embedding model, BAAI/bge-base-en-v1.5 (768 dimensions, CLS pooling, normalised; never trained in this work), maps text to the unit sphere:

$$e(x) = \frac{f(x)}{\|f(x)\|} \in S^{767} \subset \mathbb{R}^{768}.$$

A question matched against statements is embedded with the model's search instruction p in front: $e_q(x) = f(p \oplus x)/\|f(p \oplus x)\|$. Vectors are centred on a mean μ of 20,000 evenly sampled sentences, frozen before anything is stored. 1,024 directions $r_k \sim \mathcal{N}(0, I_{768})$ are drawn once (seed 0) and frozen; each bit records which side of one hyperplane the point falls on:

$$b_k(x) = \mathbf{1}[r_k \cdot (e(x) - \mu) > 0], \qquad k = 1, \ldots, 1024.$$

A fingerprint is 128 bytes; the 9M-token collection (277,681 sentences) is 35.5 MB of fingerprints. Agreement between two fingerprints is the number of equal bits,

$$A(x,y) = 1024 - \text{popcount}\left(b(x) \oplus b(y)\right), \qquad \mathbb{E}[A] = 1024\Big(1 - \frac{\theta(x,y)}{\pi}\Big),$$

because a random hyperplane separates two points with probability equal to their angle over π (Goemans and Williamson 1995; Charikar 2002). With 1,024 independent bits, Hoeffding's inequality gives $\Pr[|A/1024 - (1-\theta/\pi)| \geq \varepsilon] \leq 2e^{-2048\varepsilon^2}$: for $\varepsilon = 0.05$, the angle is read to within ±9° with probability at least 0.988. Measured: two random stored sentences share 510 bits on average (90.3°), standard deviation 34 bits.

2.2 Ranking (the vector law)

The memory holds sentences s_j (each ingested sentence is its own cue) and taught cues (c_i, j_i), a question and the sentence it points to. For a question q every sentence gets a weak score (the

question with the search instruction against the sentence) and a strong score (the best match through any cue, question against question):

$$w_j(q) = A\big(b_q(q), b(s_j)\big), \qquad \sigma_j(q) = \max\Big([s_j \text{ ingested}] \cdot w_j(q),\ \max_{i:\, j_i = j} A\big(b(q), b(c_i)\big)\Big).$$

Sentences are ordered lexicographically by (σ_j, w_j), newest first on ties; the code uses the single integer key $\sigma_j \cdot 1025 + w_j$, exact because $w_j \le 1024$. The first readout is the top 10 of this order, as bare sentences with their source files (about 41 tokens each).

2.3 Teaching: adding a point, never moving one

When the model finds an answer, it may teach the question c as a cue for the answering sentence s_j. The whole update is

$$\mathcal{C} \leftarrow \mathcal{C} \cup \{(b(c), j)\},$$

written as a new append-only shard; no stored bit, direction, mean or weight changes. Two properties follow directly from the ranking formula.

- **Integrity is exact.** For any two stored items $A'(x, y) = A(x, y)$, since neither fingerprint changed. The manifest records a SHA-256 of every shard and verifies it on load.
- **Teaching is monotone.** $\sigma'_j(q) = \max(\sigma_j(q), A(b(q), b(c)))$ and $\sigma'_{j'}(q) = \sigma_{j'}(q)$ for all $j' \neq j$: teaching can raise only the taught sentence, and can lower no other sentence's score.

Why one point generalises. A new wording q' receives the taught sentence first when $A(b(q'), b(c)) > \max_{j' \neq j} \sigma_{j'}(q')$, i.e. (in expectation) when q' lies closer to the cue than to every competing sentence. The set of such questions is the cue's region of the sphere (its Voronoi cell); every rewording inside it lands on the answer. The embedding model's contrastive training places same-meaning texts at small angles, and a question sits closer to an earlier question of the same meaning than to the statement that answers it. Measured on the 60 test facts at 9M: a second wording shares on average 692 of 1,024 bits with the first wording taught as a cue (58.4°), 641 with its own answer sentence (67.3°), and 671 with the best of 277,680 competing sentences (62.0°). The cue beats every competitor for 48 of 60 questions; the answer sentence alone does for 14.

2.4 The routing law

The routing law was authored outside this work (by the author, with an enumerative expression search) and used exactly as received. Its source is not released with this preprint; this section states what it reads, what it can serve and what it was verified to guarantee. For each question the server measures eight yes/no observations, all free (no model call), from the memory's own agreement counts and a keyword index: whether the top candidate came through a taught cue with a clear lead; whether the meaning lane's best sentence stands clear of the rest; whether it is crowded by near-ties; whether anything stands out from the collection at all; whether the question holds a rare word; whether the keyword lane's best result contains it; whether the meaning and keyword lanes agree; and whether the readout last served failed.

Six lanes serve readouts: CERTIFIED (the one taught sentence), MEANING (vector top 10), KEYWORD (BM25 top 10, SQLite FTS5), MIX (meaning top 5 + keyword top 5, deduplicated), WIDE (meaning top 20) and DIG (no readout; the model asks the memory in other words). The law turns the eight observations into an order over the lanes, serves the first, and after a failed readout moves to the next. It was verified exhaustively before use: every one of the 256 observations names a lane, no lane is served twice for one question, and every question reaches DIG within six rounds, so the loop always ends.

2.5 Self-expansion

In the self-expanding configuration the model teaches every accepted answer (the question as written, the sentence as the readout showed it). The memory therefore grows by one cue per answered question; later rewordings of those facts can be served one certified sentence. A certificate is only as good as the answer that taught it; the model reading the readout is the exact judge, and it can ignore a wrong certificate and search (§5.5).

3. Cost model

Tokens are estimated as characters / 4 from each tool server's own log of every call. Each question is priced as a fresh conversation of R rounds: round i's input is the instructions S, the question q, the first readout (for memory arms) and every tool input and output returned before it:

$$\text{tokens}(q)=\sum_{i=1}^{R}\Big(S+q+\text{readout}+\sum_{t<i}(\text{input}_t+\text{output}_t)\Big)$$

This rule (fixed before any live run) counts every logged call as ending a round, including a closing teach; we also report the CMLM with its closing teach riding the answer (no extra round), which is how the memory's own loop sends it. Price per token multiplies both arms and cancels in every ratio. Because the agent harness used here re-sends a large fixed prompt (about 40,000 tokens) with every model request, we also report *model requests per right answer*, counted from each agent's transcript.

A consequence for the readout. Write h for the share of questions whose answer is in the first readout, C_{hit} and C_{miss} for the measured cost of a question answered from it and of one that needed digging. Then tokens per question $= C_{\text{hit}} + (1-h)D$ with $D = C_{\text{miss}} - C_{\text{hit}}$ (measured: 5,122 / 2,848 / 3,027 at 1M / 3M / 9M). A readout of k bare sentences costs about $41k$ tokens. On the 184 training questions only, going from 10 to k sentences changes cost by $(k-10)\cdot 41 - (\Delta\text{hits}/184)\cdot D$: +213 / +34 / +58 for $k=5$, +38 / +81 / +90 for $k=15$, +37 / +91 / +196 for $k=20$. Ten bare sentences is the cheapest first readout at all three scales, and is the size used throughout.

4. Experimental setup

4.1 Corpus and slices

20 Newsgroups (scikit-learn distribution): 18,846 posts, 8,963,715 tokens, 277,681 sentences in the memory. Slices of about 1M, 3M and 9M tokens hold every post that answers a test question plus other posts in a fixed random order (seed 5) until the size is reached, so smaller slices nest in

larger ones. Each slice's memory is cut from the full memory (same fingerprints, directions and mean), untaught at the start of every run.

4.2 Questions

244 questions were written by a Claude agent from candidate sentences and kept only if they share no content word and no number with the answering sentence and the answer occurs in it; they were shuffled (seed 0) into 60 test and 184 training questions, one post per question. Each test fact also has a second wording written by another agent (54 of 60 share no content word with the first wording or the answer sentence). Because this construction defeats keyword search by design, a second set of *natural* queries was written by two fresh agents who never saw the memory: one search-style query per fact naming the distinctive term a searcher would know, never the answer (182/184 training and 60/60 test queries share a content word with their answer sentence).

4.3 Agents, arms and grading

Every answering agent is a fresh Claude Opus 5.5 instance working in its own folder, with the same prompt apart from the folder path; each agent answers 30 questions (two halves per scale) and may make at most 12 tool calls per question. The instructions are word-for-word identical across arms except for the section describing the tools. *Tools*: `search` (case-insensitive regex over every line, at most 40 matching lines) and `read` (up to 120 lines of a file). *Lean CMLM*: the first readout of 10 sentences with each question, then `recall` (5 bare sentences for a reworded query) and `teach`. *Router*: as lean, but the first readout is the router's and `next` gives the router's next readout. Answers were graded by fresh blind judges, 60 shuffled items each with arm labels hidden, under a fixed rule (same fact as the expected answer; extra detail allowed; judge the answer, not the cited file). Every protocol, script and input was hashed (SHA-256) before its agents ran.

5. Results

5.1 First-time questions, word-disjoint benchmark (replication)

scale	tools right	tools tok/right	lean right	lean tok/right	tools/lean	McNemar (only lean / only tools, p)
1M	56/60	5,732	56/60	4,040	1.42	2 / 2, 1.0
3M	51/60	5,329	55/60	5,884	0.91	4 / 0, 0.125
9M	53/60	8,969	52/60	8,848	1.01	3 / 4, 1.0

Table 2. Lean CMLM vs tools, 12 fresh agents, all 60 test questions. With the closing teach riding the answer the lean ratios are 1.66 / 1.08 / 1.16.

An earlier sweep on 30 of these questions had shown the lean memory about 2× cheaper at 9M; on the 30 questions never run before, it was 0.87×, so that figure did not replicate. Only 16 of 60 were answered from the first readout at 9M.

5.2 The same facts, reworded (repeat pass)

scale	answer in first readout: taught / cues ignored	tools tok/right	lean tok/right	tools/lean	right: tools / lean
1M	47 / 37	4,017	1,893	2.12	58 / 59
3M	52 / 32	4,690	1,821	2.58	57 / 58
9M	43 / 25	4,929	3,641	1.35	56 / 56

Table 3. Second wordings of the 60 facts; lean agents inherit only the memory their pass-1 agents left (8–20 cues per half); tools agents re-search. The second wordings were also easier to search: the tools agents' own cost fell to 0.73 / 0.98 / 0.58 of their pass 1.

5.3 Why the untaught advantage does not grow with scale

answer in top 10 (60 test questions)	1M	3M	9M
1,024-bit fingerprints (the memory)	33	24	17
exact 768-dimensional vectors	33	30	24
4,096-bit fingerprints (same vectors)	39	30	24
best wrong sentence's agreement (bits) / the answer's	670 / 646	678 / 648	687 / 651

Table 4. Crowding and fingerprint noise. On training questions at 9M: 1,024 bits 77/184, 4,096 bits 89/184, exact 88/184.

Two measured causes. *Crowding*: each tripling of the collection moves the best wrong sentence about 8 bits closer to the question while the answer stays put (an extreme-value effect of more candidates); exact vectors lose answers too. *Fingerprint noise*: 1,024 bits lose nothing at 1M and 7 of 60 at 9M; 4,096 bits recover the exact ranking. A keyword search does not crowd the same way, which is why the tools agents' cost rose only about 1.5× from 1M to 9M while the lean memory's doubled.

5.4 The routing law

On the word-disjoint benchmark (offline, real lanes). The law as received cost 6–33% more tokens per right answer than MEANING-then-DIG, because the keyword lane held the answer for only 0–5% of these questions and the law's keyword trigger fired on 0–3 of 60 (the law had been authored on traces simulated from a brief that assumed otherwise). A simpler law re-authored on the real training traces tied the plain memory on first-time questions and saved 4.8% / 14.1% / 7.9% on repeats.

On natural queries, the regime it was built for (offline). Every lane works (at 9M: MEANING 85%, KEYWORD 90%, MIX 92%, WIDE 88% of readouts hold the answer); the law as received is cheaper than MEANING-then-DIG in every held-out cell: novel 1,022 / 1,178 / 1,229 vs 1,316 / 1,262 / 1,369 tokens per right answer; repeats 1,004 / 875 / 1,054 vs 1,138 / 1,010 / 1,145. A fixed MIX-first policy was cheaper still at 3M and 9M.

Live, in the serving path, natural queries. The live router's first choices matched the offline test exactly.

scale	tools: right, tok/right, req/right	lean: right, tok/right, req/right	router: right, tok/right, req/right
1M	60, 1,628, 0.95	57, 1,886, 0.44	58, 1,743, 0.48

3M	60, 1,819, 1.03	57, 2,500, 0.72	58, 2,463, 0.55
9M	59, 2,558, 2.19	57, 3,308, 0.67	58, 2,595, 0.53

Table 5. 18 fresh agents, 540 blind grades. Router vs lean: 8% / 2% / 27% fewer tokens per right answer. Router vs tools: tools 7% and 26% cheaper at 1M and 3M, even at 9M; router 2.0 / 1.9 / 4.1× fewer requests.

5.5 The self-expanding CMLM vs a searching agent (main result)

A stream per scale and half: session A, the 30 natural queries; session B, the same 30 facts in their second wordings. The CMLM agent uses the router and teaches every answer; one memory per (scale, half) carries from A to B (59–60 cues after A). Tools agents start fresh in each session. After session A the router served a single certified sentence first for 39 / 36 / 32 of 60 reworded questions.

scale	segment	tools right	tools tok/right	CMLM right	CMLM tok/right (teach riding)	tools/CMLM tokens	requests ratio
1M	A (first time)	60/60	1,849	59/60	2,030 (1,065)	0.91	2.4×
	B (reworded)	60/60	3,219	59/60	1,762 (934)	1.83	4.9×
	stream	120/120	2,534	118/120	1,896 (1,000)	**1.34**	**3.5×**
3M	A	59/60	1,999	58/60	2,980 (2,006)	0.67	2.7×
	B	58/60	3,763	59/60	2,236 (1,354)	1.68	9.1×
	stream	117/120	2,874	117/120	2,605 (1,677)	**1.10**	**5.0×**
9M	A	60/60	2,270	58/60	4,156 (3,032)	0.55	1.5×
	B	56/60	6,478	59/60	3,621 (2,548)	1.79	4.5×
	stream	116/120	4,301	117/120	3,886 (2,788)	**1.11**	**3.0×**

Table 6. 24 fresh agents, 720 blind grades. Exact McNemar per session: 1M A 0/1, B 0/1; 3M A 1/2, B 2/1; 9M A 0/2, B 4/1 (only CMLM right / only tools right); all $p \geq 0.375$. With the teach riding the answer the stream ratios are 2.53 / 1.71 / 1.54.

The CMLM is cheaper over the stream at every scale, and much cheaper on the reworded half; on first-time questions it is more expensive than the searching agent by the strict rule, because each answer's teach is priced as its own round, and at 9M also with the teach riding. One certificate was wrong at all three scales (one question was served the answer sentence of another); every agent that received it rejected it, searched, and re-taught the correct sentence.

5.6 Negative and small results

- **A taught vocabulary of question shapes did not transfer.** Fresh agents wrote regex "shapes" (a question signal plus a candidate pattern) from training misses; 15 passed a training-set gate and were organised in nets of 7 under supervisor laws. On training they rescued 39 answers and lost none; on the 60 test questions they fired 30 times, rescuing 3 and losing 2. First-readout hits at 9M: 20 without, 19–20 with.
- **A hierarchical "growing net"** of clustered sub-memories with local stop rulings answered fewer questions than the flat memory (14 vs 29 at 50k; 6 vs 8 at 9M): a local "confident" verdict is not an exact judge.

- **Pseudo-relevance feedback** in bit space lowered first-readout hits by 9–11 of 184 training questions.
- **A question-side linear adapter** trained on the 184 training pairs raised held-out first-readout hits by 0–3 of 60 (9M first wordings 17 → 20); consistent in direction, not significant.
- **The single learned reordering law** in the memory never switched on (prequential 5 vs 5 after 184 examples).

6. Independent audit

A fresh agent with no prior context recomputed the replication and repeat-pass figures from the raw logs and grades with its own code: every figure matched; re-running all 1,262 search and read calls reproduced the logged token totals within 0.1%; the 360 memory readouts replayed exactly. Pooled tools/lean tokens per right answer: 1.08 [0.82–1.44] on first-time questions and 1.86 [1.39–2.77] on repeats (95% bootstrap over questions); with cache-style billing (new tokens 1.25×, re-sent 0.1×) 1.07 and 1.40; the agents' actual billed input, dominated by the harness's fixed prompt per request, favoured the memory 2.4× and 3.3× (cache-weighted). The audit named biases favouring the memory (word-disjoint questions; one sentence per answer; embedding compute not counted; no semantic tool or notes for the baseline) and noted that the repeat pass was designed after first-pass results were known.

7. Limitations and threats to validity

1. **Task.** Single-fact lookups in one domain (1993 Usenet). Multi-step research, synthesis and multi-document answers are untested.
2. **Question construction.** The benchmark questions share no content word with their answers, which handicaps keyword search; the natural queries (§5.4–5.5) partly address this and change the picture (keyword search becomes strong).
3. **Same model family throughout.** Question writers, answering agents and judges are all Claude; correlated biases cannot be excluded. Judges split on a few ambiguous items; we checked grading consistency across arms for each disputed item and report it.
4. **Sample size.** 60 questions per scale; accuracy differences are not significant; per-scale token ratios have wide intervals (about ±40% by bootstrap in the audit), and agent-to-agent variation is large.
5. **Pricing.** Characters / 4 is an approximation; the per-round rule over-counts a closing teach (we report both); request counts depend on this harness's fixed prompt.
6. **Baselines.** No tools-plus-notes agent, no semantic search for the baseline, no hybrid retrieval baseline; these are the natural next comparisons.
7. **Scale.** Tested to 9M tokens; the untaught memory crowds with size (§5.3). 1B tokens is untested.
8. **Deviations** (Appendix B) were small and are counted in the figures.

8. Code and data availability

The memory package, the tool servers, the experiment and grading scripts, every agent's instructions, questions and answers, every tool server's call log, the blind judges' verdicts, the pre-registered protocols and their SHA-256 hashes are available from the author and will be released with the published version. Each reported table is regenerated from the logs and verdicts by the script that produced it. The routing law is not released. Every answering agent, question writer and judge was Claude Opus 5.5.

9. Conclusion

A fingerprint memory that answers first matches a searching agent's accuracy on single-fact questions. Its token advantage over that agent is modest on first-time questions and depends on the regime: large when keyword search is handicapped and the collection is small, absent when queries share words with their answers. Where the memory is clearly better is repetition: an append-only store of taught questions generalises to new wordings through the embedding's geometry, is served as a single certified sentence by the routing law, and makes reworded research 1.7–1.8× cheaper in tokens and 4.5–9× cheaper in requests while never altering what is stored. The cost of that growth is paid when the facts are first found. The measured obstacles to larger collections, crowding and fingerprint noise, suggest the next steps: longer fingerprints, a keyword lane beside the meaning lane, and baselines that also keep notes.

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AI assistance. Code, experiment orchestration and much of this text were produced with Claude (Anthropic) under the author's direction; the answering agents, question writers and judges were separate Claude Opus 5.5 instances, as described in §4.

Appendix A. The routing law

The routing law's source is not released with this preprint. §2.4 states its inputs, its six lanes and the properties it was verified to have; §5.4 and §5.5 report how it performed.

Appendix B. Protocol deviations

- Replication: four questions reached 13 calls (limit 12); one tools agent piped a search through a local filter once (unused); one lean agent found an answer through another question's recall (graded incorrect anyway). Judges split on one ambiguous item; graded consistently, the tools arm loses one answer at 1M and 9M.
- Teaching-cycle experiment: the operator mistyped the teachers' batch lists (question labels only; no held-out content reached a teacher; recorded before scoring); one teacher emptied its shapes' taught-from lists (a strict-gate sensitivity run is reported).
- Stream: two CMLM questions reached 13 calls counting their required teach; one tools agent piped a search through a local filter once; the one wrong certificate, at all three scales, was caught each time.
- One test question is ambiguous (two posts carry different counts); the memory ranks the other post first; grading was consistent across arms (every answer leading with the other count graded incorrect).

Appendix C. The geometry of a taught shortcut

This appendix collects the derivation of §2.3 in one place. Its central point: teaching does not adjust any point. The memory is append-only; a taught shortcut adds one new 1,024-bit fingerprint linked to the sentence that answered, and every existing fingerprint stays identical bit for bit. It generalises because of where the new point sits: a point in a meaning space claims the region of all questions closer to it than to anything else.

C.1 The pipeline, step by step

1. **Sentence to point.** $e(x) = f(x)/\|f(x)\| \in S^{767}$; questions against statements use $e_q(x) = f(p \oplus x)/\|f(p \oplus x)\|$; centring on μ, the mean of 20,000 sentences, frozen before anything is stored.
2. **Point to fingerprint.** $r_k \overset{\text{iid}}{\sim} \mathcal{N}(0, I_{768})$, $b_k(x) = \mathbf{1}[r_k \cdot (e(x) - \mu) > 0]$, $k = 1..1024$: 128 bytes; 35.5 MB for 277,681 sentences.
3. **Bits measure angles.** $\Pr[b_k(x) \neq b_k(y)] = \theta/\pi$; $A = 1024 - \text{popcount}(b(x) \oplus b(y))$, $\mathbb{E}[A] = 1024(1 - \theta/\pi)$; Hoeffding: within ±9° with probability ≥ 0.988. Random pairs: 510 bits (90.3°), sd 34.
4. **What the memory holds.** $\mathcal{F} = \{(b(s_j), s_j)\}$ and $\mathcal{C} = \{(b(c_i), j_i)\}$; a taught cue is measured without the search instruction, since it is a question compared with questions.
5. **Ranking.** $w_j(q) = A(b_q(q), b(s_j))$, $\sigma_j(q) = \max([s_j \text{ ingested}]\, w_j(q), \max_{i:j_i=j} A(b(q), b(c_i)))$; key $\sigma_j \cdot 1025 + w_j$.
6. **Teaching.** $\mathcal{C} \leftarrow \mathcal{C} \cup \{(b(c), j)\}$; $\sigma'_j = \max(\sigma_j, A(b(q), b(c)))$, $\sigma'_{j'} = \sigma_{j'}$ for $j' \neq j$, $w' = w$. Integrity: $A'(x, y) = A(x, y)$ for all stored pairs. Monotone: only the taught sentence can rise.
7. **Region.** q' gets the taught answer first iff $A(b(q'), b(c)) > \max_{j' \neq j} \sigma_{j'}(q')$, i.e. $\theta(q', c) < \min_{j' \neq j} \theta(q', s_{j'})$ in expectation: the cue's Voronoi cell.

C.2 Measured geometry (60 test facts, 9M memory)

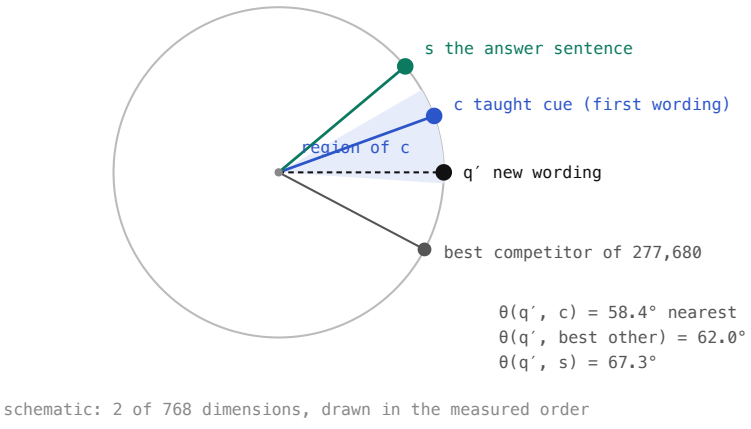


Figure C1. Schematic (2 of 768 dimensions), drawn in the measured order. A second wording sits at 58.4° from its taught cue (692 of 1,024 shared bits), 62.0° from the best of 277,680 competing sentences (671), and 67.3°

from its own answer sentence (641). The cue beats every competitor for 48 of 60 questions; the answer sentence alone for 14.

Live. With the cues the first-pass agents actually taught, the answer was in the first readout of 10 for 43 of 60 reworded questions at 9M against 25 with those cues ignored (52 vs 32 at 3M, 47 vs 37 at 1M). **Where it can fail.** A cue also claims nearby questions that mean something else; measured interference was small (0 and -2 of 60 when 184 unrelated cues were taught), and the average margin is tight (21 bits against about 15 bits of per-pair spread), which is why 12 of 60 still lose.

C.3 Two ways to generalise: derivation and proximity

	fluidnet (by derivation)	**CMLM (by proximity)**
what is taught	a class: a signal (regex over the input), an applier that derives the fix from the line in front of it, a property	one point in a meaning space: a question linked to the sentence that answered it
judge	the test suite, exact	the model reading the readout, exact; the memory only proposes
model involved	none	arithmetic model-free; geometry from a frozen embedding model
measured	240 of 240 unseen instances of taught fault shapes (code; reported for the fluidnet repair system; not measured here)	18 more first-readout answers in 60 at 9M (open text)

Merged, the loop is: proximity narrows millions of sentences to the ten worth reading, derivation and the exact judge turn the right one into a proven answer, and each accepted answer is taught back as one more point. Tested inside the CMLM, fluidnet-style classes over open-text questions rescued 3 and lost 2 of 60 held-out questions (§5.6): on open text, proximity does the generalising and derivation's part is the exact judge.

C.4 What never happens

- No gradient step, no weight update, no stored point moved: forgetting what was stored is impossible by construction.
- The optional learned reordering law (one of $3^8 = 6{,}561$ lane settings over the top 8) never switched on in these tests.
- The trained question-side adapter moves only the question, never the stored fingerprints (+0 to 3 first-readout answers in 60).